ML Model Evaluation & Explainability Report

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1. Dataset & Problem Detection Summary

Dataset File:	loan_data.csv	**Total Rows:**	5004
Target Column:	Loan_Status	**Total Columns:**	16
Problem Type:	Classification	**Test Split:**	20%

2. Preprocessing & Validation Protocol

Numerical Imputer	Median	**Categorical Encoding**	Onehot
Categorical Imputer	Most_frequent	**Feature Scaling**	Standard
Random State	42	**CV Protocol**	3 Folds

3. Model Performance Comparison

Rank	Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC	Train Time (s)
1	Logistic Regression	0.8821	0.7911	0.7149	0.7511	0.9412	0.37
2	Decision Tree	0.8192	0.6269	0.6747	0.6499	0.7709	0.23
3	Random Forest	0.8561	0.7612	0.6145	0.6800	0.9126	4.49

4. Recommended Champion Model

Recommended Model: Logistic Regression
Criterion: Highest F1 Score (0.7511).
Rationale: Superior performance and fold consistency on held-out validation data.

5. Global Feature Importance (SHAP)

Feature	Mean Absolute SHAP Value	Relative Impact
Credit_Score	2.0776	100.0%
Income	1.5902	76.5%
Employment_Type_Salaried	1.2125	58.4%
Experience	1.1152	53.7%
Existing_Loans	1.1054	53.2%
Loan_Amount	0.9686	46.6%
Employment_Type_Unemployed	0.5803	27.9%

Employment_Type_Self-Employed	0.3418	16.5%